



Functional and Non-Functional Requirements - RabbitCTF

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1 Functional Requirements

FR-1: User management

- **FR-1.1: Register users:** The system must allow new users to register using a form that requests a username, email address, and password, for the purpose of creating unique and persistent accounts to access the platform and participate in the CTF competition. The system must validate that all required fields are complete and that the email address and username are unique before creating the account.
- **FR-1.2: User login:** The system must allow users and administrators to log in by validating their credentials, consisting of a username and password, establishing authenticated access that enables views and permissions according to the assigned role. The session must remain active until the user manually closes it, the event ends, or the administrator revokes access.
- **FR-1.3: Register initial system administrator:** The system must allow, only during the first run when there are no registered administrators, the creation of an initial administrator using a form that requests a username, email address, and password. Once the first administrator has been registered, the system must permanently disable this option to prevent new initial administrators from being registered in this way.
- **FR-1.4 – Create teams:** The system must allow an authenticated user who does not belong to any team to create a new team using a form that requests a name and password. The system must validate that the team name is unique, automatically register the creator as captain, and enable the team so that other users can join using the defined password.
- **FR-1.5 – Consult teams:** The system must allow authenticated users to consult a general list of registered teams, showing the name and current number of members for each team.
- **FR-1.6 – Viewing information about one's own team:** The system must allow an authenticated user who belongs to a team to view detailed information about their team, including the captain's name, the complete list of members, and the total score accumulated in the competition. If the user does not belong to any team, the system must notify them of this condition by means of an informative message visible in the same section, which will remain active until the user closes the notification or joins a team.

- **FR-1.7 – Joining teams:** The system must allow an authenticated user to join an existing team by entering the team name and access password. The system must validate that the user does not currently belong to any team, that the password provided is correct, and that the team has not reached the maximum number of members allowed before completing the addition.
- **FR-1.8 – Deleting teams:** The system must allow a captain or administrator to delete a team, subject to explicit confirmation of the action. When deleting the team, the system must also delete all entries and scores associated with the team and its members.
- **FR-1.9 – Request password recovery:** The system must allow a user who has forgotten their password to request that the administrator reset their account using a form available on the login interface. The system must generate and record the request, notifying the user that their request has been sent to the administrator and keeping this notification visible until the user closes it manually.
- **FR-1.10 – Reset user password:** The system must allow the administrator to reset a user's password from the administration panel, either by setting a new password or generating a temporary password that allows the user to access their account again.
- **FR-1.11 – Require temporary password change:** The system must require any user with a temporary password assigned by the administrator to change their password on their next login, preventing access to other features until the new password is successfully registered.

FR-2: Team management

- **FR-2.1 – Create challenges:** The administrator must be able to create a new challenge by entering its main information, including title, description, category, tags, difficulty level, score, scoring mode, and operational data such as URL or port when applicable, leaving the challenge initially in draft status or ready for publication according to the established configuration.
- **FR-2.2 – Configure challenge visibility:** The administrator must be able to define the visibility of the challenge at the time of its creation or editing, establishing whether it will be publicly available to competitors or remain in draft status until its manual publication.

- **FR-2.3 – Edit existing challenges:** The administrator must be able to modify any attribute of an existing challenge without affecting the history of flag submissions or previously recorded score references.
- **FR-2.4 – Manage flag and sensitivity:** The administrator must be able to set the flag associated with a challenge and define whether its validation will be case-sensitive or not.
- **FR-2.5 – Configure scoring mode:** The administrator must be able to assign the scoring mode for the challenge, choosing between a static or dynamic value, and in the latter case, define the decay factor and the minimum achievable score.
- **FR-2.6 – Attach files to the challenge:** The administrator must be able to attach one or more files to the challenge during its creation or editing, ensuring that the files are linked to the challenge and available to competitors according to the configured visibility.
- **FR-2.7 – Delete challenges with explicit confirmation:** The administrator must be able to delete a challenge after explicitly confirming the action, ensuring that the deletion is only executed after such confirmation.
- **FR-2.8 – Ensure consistency after challenge deletion:** The system must delete all attached resources and accessible references associated with the challenge, maintaining consistency with historical records in accordance with the established retention policy.
- **FR-2.9 – Attach files to challenges:** The administrator must be able to attach one or more files to a challenge during its creation or editing, respecting a maximum individual size of 100 MB and a cumulative limit of 500 MB per challenge.
- **FR-2.10 – Restrict allowed file types:** The system must only accept files with the extensions .zip, .tar.gz, .txt, .pdf, .pcap, .png, or .jpg; executable files (.exe, .sh, .bat) should only be allowed when contained within accepted compressed files.
- **FR-2.11 – Store files associated with the challenge:** The system must store associated files in such a way that they are referenced to the corresponding challenge and available for download only by authenticated competitors who have access to the challenge.
- **FR-2.12 – Delete files associated with the challenge:** The administrator must be able to delete files previously attached to a challenge, ensuring that they are removed from both storage and system references.
- **FR-2.13 – Create challenge categories:** The administrator must be able to create new challenge categories to facilitate their classification and filtering within the system.

- **FR-2.14 – Edit challenge categories:** The administrator must be able to modify the name or description of an existing category without affecting the association of challenges already classified under it.
- **FR-2.15 – Deactivate challenge categories:** The administrator must be able to deactivate a challenge category, preventing it from being used in new creations or edits, without removing it from the system or affecting previously classified challenges.
- **FR-2.16 – Delete challenge categories:** The administrator must be able to delete a challenge category only when there are no active challenges associated with it, ensuring the integrity of references.
- **FR-2.17 – Change challenge visibility:** The administrator must be able to toggle the visibility of a challenge between visible and invisible to competitors, without modifying its internal configuration or affecting the history of existing submissions.
- **FR-2.18 – Define challenge scoring mode:** The administrator must be able to set the scoring mode of a challenge as static or dynamic, according to the parameters defined for the event.
- **FR-2.19 – Configure dynamic scoring parameters:** The administrator must be able to define the decay factor and the minimum achievable score in dynamic scoring mode, so that the value of the challenge decreases as the number of registered solutions increases, without falling below the configured minimum.
- **FR-2.20 – Display challenge solution history:** The system must display a solution history on the challenge tab that includes, at a minimum, the user or team that solved it and the date and time of the solution.
- **FR-2.21 – Allow administrative access to complete history:** The system must allow administrators to always view the complete history of solutions for a challenge, without restrictions due to event policy.
- **FR-2.22 – Restrict visibility of history according to policy:** The system must allow competitors to view the history of solutions for a challenge only when authorized by event policy.

FR-3: Participation and validation

- **FR-3.1 – Accessing the challenge panel:** The authenticated competitor must be able to access a panel showing all available challenges for the current event to which their team has access.
- **FR-3.2 – Display key information for each challenge:** For each available challenge, the system must display its title, category, difficulty level, current score, and current status with respect to the user's team (solved or unsolved) on the dashboard.
- **FR-3.3 – Submit challenge flag:** The competitor must be able to submit a text string as a flag for a published challenge using the corresponding form. The system must validate the flag and return the validation result within a maximum of 2 seconds without exposing sensitive information.
- **FR-3.4 – Confirm correct flag:** When the flag sent matches the configured solution, the system must display a non-blocking confirmation message on the corresponding challenge interface, indicating success and disappearing automatically within a maximum of 8 seconds. The system must mark the challenge as solved for the corresponding team and update the team and user scores on the interface within a maximum of 5 seconds.
- **FR-3.5 – Notify incorrect flag:** When the flag sent does not match the configured solution, the system must display a generic non-blocking message on the corresponding challenge interface, indicating an error without revealing details or patterns of the solution. The message must remain visible for 8 seconds before disappearing automatically.
- **FR-3.6 – Log validation attempts:** The system must log each flag submission attempt, including user or team, challenge, date and time in UTC format, and validation result. The flag value must not be stored in plain text in any log.
- **FR-3.7 – Apply validation settings:** The system must apply flag validation according to the configuration established for each challenge, considering case sensitivity and the scoring rules defined for that challenge.
- **FR-3.8 – View validation history:** The administrator must be able to view the complete validation history from the administrative panel, with filtering options by user, team, challenge, and date range. The system must return results in a maximum of 3 seconds for queries of up to 10,000 records.

- **FR-3.9 – Display validation statistics:** The administrator must be able to view aggregate statistics in the administrative panel, including the success rate per challenge, the average number of attempts before resolution, and the list of challenges with the highest number of failed attempts. The system must calculate and display the updated values within a maximum of 10 seconds after the request.
- **FR-3.10 – Apply challenge score:** The system must automatically calculate and apply the corresponding score when a flag is validated as correct, according to the scoring mode configured in the challenge (static or dynamic).
- **FR-3.11 – Apply score in static mode:** The system must assign exactly the score value configured for the challenge when it is in static mode, reflecting the result in the team's score and, when applicable, the user's score, within a maximum of 5 seconds after validation.
- **FR-3.12 – Apply scoring in dynamic mode:** The system must calculate the score in dynamic mode using the decay factor and minimum score configured, based on the number of solutions registered at the time of validation, and reflect the resulting score on the team and, when applicable, on the user, within a maximum of 5 seconds after validation.
- **FR-3.13 – Maintain consistency of scores:** The system must ensure that the scores applied are consistent with the leaderboard and that the values awarded in previous validations are not modified, applying the calculation only to the current validation event.
- **FR-3.14 – Mark challenge as resolved for the team:** The system must automatically mark a challenge as “resolved” for all team members when any member has correctly validated the corresponding flag, ensuring that the status is visible to the entire team within a maximum of 5 seconds after validation.
- **FR-3.15 – Maintain access to resolved challenge:** The system must allow all team members, even when a challenge is in “resolved” status, to access its description, download associated files, and copy the challenge's operational data without restrictions.
- **FR-3.16 – Update challenge status after administrative reversal:** The system must automatically update the status of a challenge to “unresolved” for the team when a previously accepted validation is reversed by an administrative action, reflecting the change in scores and the leaderboard within a maximum of 10 seconds.

- **FR-3.17 – Provide successful feedback after flag validation:** The system must display a visible confirmation message to the competitor when the submitted flag is successfully validated, updating the challenge status and corresponding score within a maximum of 5 seconds after validation.
- **FR-3.18 – Provide error feedback on flag validation:** The system must display a generic error message to the competitor when the submitted flag is invalid, without revealing clues, patterns, or fragments of the correct flag, maintaining the confidentiality of the solution.
- **FR-3.19 – Allow navigation after validation feedback:** The system must allow the competitor to continue browsing other challenges or resubmit the flag immediately after receiving any type of feedback, without blocking interaction with the platform.
- **FR-3.20 – Display solution history allowed by policy:** The system must allow the authenticated competitor to view a visible history of solutions from the challenge tab only when authorized by the event policy, displaying only the allowed fields and excluding any flags or sensitive data.
- **FR-3.21 – Allow administrative access to complete history:** The system must allow the administrator to always view the complete history of solutions for a challenge, regardless of the restrictions established by the event policy.
- **FR-3.22 – Update solution history in real time:** The system must update the history of solutions visible to authorized users within a maximum of 10 seconds after a new solution is registered, maintaining consistency with the status of the challenge and the associated scores.
- **FR-3.23 – Allow copying of visible challenge operational data:** The system must allow authenticated competitors to copy visible challenge operational data, such as addresses or short commands, to the clipboard, ensuring that the copied content corresponds exactly to the displayed text and does not include flags or sensitive information.
- **FR-3.24 – Allow administrative copying of operational data:** The system must allow the administrator to copy the same operational data from their administrative view, without additional restrictions and with equal accuracy in the copied content.
- **FR-3.25 – Restrict copying of operational data in non-visible challenges:** The system must block the action of copying operational data from the challenge in the competitor's

view when the challenge is not visible according to the event policy, keeping the action available only to the administrator.

- **FR-3.26 – Visually confirm the copy action:** The system must display a visible confirmation to the user (competitor or administrator) immediately after executing the copy action, remaining visible for at least 2 seconds without blocking interaction with the platform.
- **FR-3.27 – Apply default flag submission limits:** The system must allow, by default, a maximum of 5 flag submission attempts every 60 seconds per user, automatically blocking new submissions for 5 minutes when this limit is reached.
- **FR-3.28 – Count attempts and apply temporary blocking:** The system must count flag submissions per user or team when the limitation policy is active, and automatically block new submissions when the configured threshold is reached, restoring permission once the blocking time has expired.
- **FR-3.29 – Notify block due to submission limit:** The system must display a clear message to the competitor when flag submissions are blocked due to exceeding the configured limit, indicating the reason for the block and a countdown timer showing the time remaining until the next permitted attempt.
- **FR-3.30 – Allow administrator to set limits:** The system must allow the administrator to set the thresholds for the attempt limitation policy with the following valid ranges: between 1 and 20 attempts allowed, a time window between 10 and 300 seconds, and a temporary block between 1 and 60 minutes.
- **FR-3.31 – Reset submissions after lockout expires:** The system must allow the competitor to resume submitting flags normally once the time restriction has expired or the attempt counter has been reset in accordance with the active policy.
- **FR-3.32 – Logging of blocking events due to submission limits:** The system must log each blocking event due to the submission limitation policy, storing only the timestamp, the user or team, and the challenge affected, without saving flags in plain text or exposing sensitive information.
- **FR-3.33 – Filter challenges by category:** The authenticated competitor must be able to filter the challenges visible on the panel according to the selected category, so that only challenges belonging to that category are displayed.

- **FR-3.34 – Filter challenges by difficulty level:** The authenticated competitor must be able to filter the challenges visible on the panel according to the selected difficulty level, displaying only the challenges that correspond to the chosen level.
- **FR-3.35 – Filter challenges by resolution status:** The authenticated competitor must be able to filter the challenges visible on the panel according to their status with respect to the team (resolved or unresolved), allowing them to focus on pending or completed challenges.
- **FR-3.36 – Filter challenges by tags:** The authenticated competitor must be able to apply filters by tags to display only the challenges that contain the selected tags.
- **FR-3.37 – Sort challenges by score:** The authenticated competitor must be able to sort the challenges visible on the panel by current score, in ascending or descending order.
- **FR-3.38 – Sort challenges by difficulty:** The authenticated competitor must be able to sort the challenges visible on the panel by difficulty level, in ascending or descending order.

FR-4: Rankings and statistics

- **FR-4.1 – View overall leaderboard:** The system must allow participants and administrators to view an event leaderboard showing teams with their accumulated scores, number of challenges solved, and sorted from highest to lowest score.
- **FR-4.2 – View Participant and Team Counts**

The system shall provide administrators with a view of the total number of registered teams and the total number of registered participants. This view must allow these aggregate counts to be filtered by a date range (e.g., registrations within a specific period).
- **FR-4.3 – View Challenge Completion Statistics by Category**

The system shall provide administrators with a view of challenge completion statistics, displaying the number and percentage of challenges solved, broken down by challenge category (e.g., Web, Crypto, Forensics). This view must allow these statistics to be filtered by challenge category.
- **FR-4.4 – View Overall Event Completion Percentage**

The system shall provide administrators with a view of the global event progress, represented by an overall percentage of challenges completed out of the total available..

- **FR-4.5 – View basic administrative statistics for the event:** The system must allow administrators to view a dashboard with event indicators that shows, in real time, the first to solve each challenge (first blood) and the outstanding player (MVP) calculated according to the highest individual score accumulated and the number of challenges solved.

FR-5: Communications and notifications

- **FR-5.1 – Display basic internal notifications on the platform:** The system must display relevant CTF events in an internal notifications section (e.g., publication of new challenges, announcements from the organizer).

- **FR-5.2 – Manage Notification Types**

The system shall allow administrators to create, edit, and deactivate distinct types of internal notifications (e.g., "Challenge Published," "Organizer Announcement," "CTF Event Started"). Each type must be defined by a unique, human-readable name and a system-wide active/inactive status.

- **FR-5.3 – Define Notification Display Format**

The system shall allow administrators to define a basic display format for each notification type. This configuration must include a title template and a visual style (e.g., color code, icon) to ensure consistency for all notifications of the same type.

- **FR-5.4 – Enforce Safe Content Policy**

The system shall implement a validation rule to prevent the inclusion of sensitive data (specifically challenge flags) in the title or message body of any notification before it is saved or dispatched.

- **FR-5.5 – Optional integration with Discord via local configuration:** The system must allow the integration of a Discord webhook to publish basic CTF announcements (e.g., new challenges and featured captures), without exposing sensitive data, provided that the organizer enables it.

FR-6: Event configuration

- **FR-6.1 – Define CTF start and end date/time:** The system must allow administrators to define the start and end date and time of the CTF event, reflecting the CTF statuses in the interface.
- **FR-6.2 – Change event status:** The administrator must be able to change the event status between “not started,” “active,” and “finished” from the administrative interface, with said status being recorded in the database and available to all users within a maximum of 5 seconds after the modification.
- **FR-6.3 – Reflect event status in the interface:** The system must display the current status of the event (“not started,” “active,” or “finished”) in a visible area of the main interface for all users, updating automatically within a maximum of 5 seconds after any status change.
- **FR-6.4 – Restrict the sending of flags according to the status of the event:** The system must block the sending of flags to competitors when the status of the event is “not started” or “finished,” and immediately enable it when the status is “active.”
- **FR-6.5 – Display an informational message when restricting the sending of flags:** The system must display a non-blocking informational message at the top of the competitors' interface when flag sending is restricted by the event status (“not started” or “finished”); the message must appear at the time of the sending attempt, remain visible for 5 seconds or until the user performs another action, and then disappear automatically.
- **FR-6.6 – Maintain the integrity of historical event data:** The system must ensure that any change in the status of the event does not modify the scores or the history of previously recorded submissions, preserving the integrity of the data at all times.
- **FR-6.7 – Create and edit event rules and guidelines:** The administrator must be able to create and edit the content of event rules and guidelines using a text editor with support for basic Markdown syntax (headings, lists, bold, italics, links, and code blocks) and a maximum limit of 10,000 characters per document.
- **FR-6.8 – Display character counter in the editor:** The system must display a real-time counter within the text editor that indicates the number of characters used out of the 10,000 available.

- **FR-6.9 – Restrict saving by character limit:** The system must prevent content from being saved when it exceeds the 10,000-character limit and display a non-blocking informational message at the top of the editor indicating that the limit has been exceeded; the message must remain visible for 5 seconds or until the user modifies the text.
- **FR-6.10 – Log versions of edited content:** The system must automatically create a new version in the history each time an administrator saves changes to the rules or guidelines, recording the date, time, and name of the administrator, along with a complete copy of the content.
- **FR-6.11 – Maintain a limited version history:** The system must retain only the last 5 versions of the rules and guidelines content; when a sixth version is generated, the oldest must be automatically deleted.
- **FR-6.12 – View version history:** The administrator must be able to view the content version history from a dedicated panel, displaying the date, time, and author corresponding to each version.
- **FR-6.13 – Preview previous versions:** The administrator must be able to preview any previous version of the content from the history panel, displaying the text with Markdown formatting rendered correctly.
- **FR-6.14 – Restore previous versions:** The administrator must be able to restore any previous version of the content by explicit confirmation; such restoration must create a new current version recorded in the history.
- **FR-6.15 – Render content visible to participants:** The system must render the content of rules and guidelines in a dedicated view accessible to all participants, applying a rendering library that processes basic Markdown and blocks any embedded HTML code or malicious scripts.
- **FR-6.16 – Reflect changes in the public view:** The system must update the content visible in the participants' public view within a maximum of 5 seconds after an administrator confirms and saves the changes.
- **FR-6.17 – Define the maximum team size:** The administrator must be able to configure the maximum size allowed per team from the event administration view, storing the value in the database and applying it automatically to team creation and joining validations.

- **FR-6.18 – Restrict the creation of or joining to teams that exceed the limit:** The system must prevent the creation of new teams or the joining of competitors to existing teams when they reach the configured maximum size, displaying a non-blocking information message at the top of the interface for 5 seconds or until the user performs another action.
- **FR-6.19 – Configure global policy for attempts or flag submission rate:** The administrator must be able to enable and define a global policy that limits the maximum number of attempts or the rate of flag submissions per team or competitor (e.g., maximum of 1 attempt every 10 seconds), which must be stored and automatically applied to all flag submissions.
- **FR-6.20 – Notify when the flag sending limit is reached:** The system must display a non-blocking informational message at the top of the competitors' interface when the configured attempt limit or rate is reached, indicating the restriction and the time remaining before the next allowed attempt; the message must remain visible for 5 seconds or until the user performs another action.
- **FR-6.21 – Apply configuration changes immediately:** The system must immediately apply any changes to global limits (maximum team size or flag submission policy) to all new operations after the modification, without altering historical records or operations already performed.
- **FR-6.22 – Perform event reset according to defined policy:** The administrator must be able to perform a complete or partial reset of the event according to a predefined policy that determines which data will be deleted (e.g., submissions and scores) and which will be retained (e.g., users, teams, and challenges), applying this policy automatically after explicit confirmation.
- **FR-6.23 – Display summary prior to reset:** Before executing the reset, the system must present the administrator with a detailed summary of the scope of the operation, indicating the data that will be deleted and that which will be retained, requiring explicit confirmation of a destructive nature in order to proceed.
- **FR-6.24 – Maintain data consistency after restart:** The system must ensure that, once the restart is complete, the leaderboard, submissions, and “resolved/unresolved” statuses

of the challenges remain consistent with the applied policy, without retaining residual information from the deleted data.

- **FR-6.25 – Avoid partial changes in case of error:** The system must execute the restart process only if all operations defined in the policy can be completed in full; in case of failure, no partial changes must be applied and a blocking informational message must be displayed to the administrator indicating the reason for the error.

FR-7: Maintenance and Deployment

- **FR-7.1 – Provide local deployment using Docker Compose:** The system must provide a local deployment procedure based on Docker Compose that allows frontend, backend, and database services to be launched, ensuring their joint execution in a local environment without requiring cloud dependencies.
- **FR-7.2 – Include orchestration file with minimal configuration:** The system must include a Docker Compose orchestration file that defines the necessary configuration of services, minimal environment variables, and persistent volumes for the database and challenge files.
- **FR-7.3 – Allow deployment with minimal commands:** The administrator must be able to start the entire system environment in a local environment using a maximum of two consecutive commands in the terminal, with the platform accessible from a modern supported browser within a maximum of 60 seconds after execution.
- **FR-7.4 – Retain persistent data between executions:** The system must maintain the data volumes and challenge files configured in Docker Compose persistently between executions, ensuring that re-execution of the deployment does not delete previously stored information.
- **FR-7.5 – Maintain persistent volumes for database and challenge files:** The system must maintain separate persistent volumes for the database and for files associated with challenges, ensuring that information is preserved between environment restarts and local deployment updates.
- **FR-7.6 – Define separate volumes in deployment orchestration:** The Docker Compose orchestration file must explicitly define two separate persistent volumes (one for the

database and one for challenge files), specifying their storage paths in the deployment configuration.

- **FR-7.7 – Preserve data after service restart:** The system must ensure that when any of the environment's services (frontend, backend, or database) are restarted, the data stored in the persistent volumes remains intact and accessible, without requiring manual intervention by the administrator.
- **FR-7.8 – Allow volumes to be relocated through configuration:** The administrator must be able to modify the paths of persistent volumes using documented configuration parameters, maintaining the integrity of existing data during the relocation.
- **FR-7.9 – Update the environment without losing persistent data:** The administrator must be able to update the local system environment—including container images and minimum environment variables—without deleting or modifying the information stored in persistent database volumes and challenge files.
- **FR-7.10 – Provide a documented update procedure:** The system must include a documented procedure indicating the steps necessary to stop services, update images, and restart the environment, ensuring the integrity of data and files during the process.
- **FR-7.11 – Restart environment after update while preserving configuration:** The system must start correctly after an update within a maximum of 60 seconds, preserving the configuration, scores, and statuses of previous challenges, without generating inconsistencies or loss of information.
- **FR-7.12 – Manage schema migrations in a controlled manner:** The system must detect when an update requires a database schema migration and explicitly indicate this requirement in the update procedure; the migration must be executed atomically and securely or, otherwise, blocked by displaying a blocking information message to the administrator with the cause of the block.
- **FR-7.13 – View basic service status:** The administrator must be able to view, from the administrative interface, the basic status of each main system service (frontend, backend, and database), showing whether they are “running” or “stopped,” with an automatic update every 10 seconds.
- **FR-7.14 – View sensitive operational logs:** The administrator must be able to access recent operational logs for each service from the administrative interface, with automatic

pagination or truncation that limits the query to the last 1,000 records per service, without affecting overall system performance.

- **FR-7.15 – Ensure non-intrusive status and log queries:** The system must ensure that service status and administrative log query operations do not block or degrade normal user operations or environment performance.

2 Non-Functional Requirements

NFR-1: Performance

- **NFR-1.1 – Initial load time:** The system must load the fully interactive main interface in a maximum of 15 seconds under local deployment conditions.
- **NFR-1.2 – Switching between views:** The system must transition between internal views in a maximum of 2 seconds under normal local network conditions.
- **NFR-1.3 – Critical operations:** Flag validation and score update operations must be completed within a maximum of 3 seconds.
- **NFR-1.4 – Stability under moderate load:** The system must maintain stable performance with no noticeable degradation in response times or interface fluidity under loads of up to 100 concurrent users in a local or medium-scale environment.

NFR-2: Real-time updating

- **NFR-2.1 – Automatic leaderboard updating:** The system must update the leaderboard automatically through real-time communication, reflecting changes within a maximum of 5 seconds of their registration.
- **NFR-2.2 – Recovery from connection loss:** In the event of a temporary connection interruption, the system must automatically resume synchronization when service is restored, ensuring the consistency of the data displayed.

NFR-3: Security

- **NFR-3.1 – Secure authentication:** The system must implement session-based authentication with encrypted passwords and session tokens with controlled expiration.
- **NFR-3.2 – Protection against failed attempts:** The system must temporarily block an account after five (5) consecutive failed authentication attempts for a period of 10 minutes.
- **NFR-3.3 – Input validation:** The system must implement validations that prevent SQL injection, XSS, and CSRF attacks on all user inputs.
- **NFR-3.4 – Encryption of sensitive data:** Credentials and flags must be stored encrypted on the server and never transmitted unencrypted.

- **NFR-3.5 – Database access control:** The database must restrict access through roles and limited permissions based on the user's role or service.
- **NFR-3.6 – Frontend exposure:** Internal configurations and validation parameters must not be accessible or visible from the client.

NFR-4: Usability and design

- **NFR-4.1 – Browser compatibility:** The interface must be fully functional in Google Chrome, Microsoft Edge, and Mozilla Firefox in current desktop versions.
- **NFR-4.2 – Function accessibility:** The user must be able to access the main functions (challenges, flags, rankings, and statistics) in a maximum of 3 clicks from the home page.
- **NFR-4.3 – Immediate feedback:** Each user action must generate a visual, textual, or auditory response within a maximum of 1 second.
- **NFR-4.4 – Interface adaptability:** The interface must retain its structure and functionality at resolutions between 1366×768 px and 3840×2160 px (4K), without overlapping or loss of visible content.

NFR-5: Deployment and reproducibility

- **NFR-5.1 – Deployment using Docker Compose:** The system must be deployable using Docker Compose, including three independent services: frontend, backend, and database.
- **NFR-5.2 – Data persistence:** The backend and database must have persistent volumes that preserve files and records between sessions.
- **NFR-5.3 – Deployment time:** The entire installation and initial execution process must not exceed 15 minutes for a user with basic technical knowledge.

NFR-6: External integration

- **NFR-6.1 – Discord integration:** The system must communicate with a Discord bot via API to send CTF event notifications (achievements, first bloods, administrative messages, and status changes).
- **NFR-6.2 – Notification latency:** Notifications must be sent within a maximum of 10 seconds of the triggering event.

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- **NFR-6.3 – Integration error handling:** The system must log communication errors with Discord and continue to function without interruption.
 - **NFR-6.4 – Token security:** Bot authentication tokens must be stored encrypted on the server and never be visible from the client.